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Test-enhanced learning in third-grade children

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Several recent studies have shown that retrieval is more efficient than restudy in enhancing the long-term retention of memories. However, studies investigating this effect in children are still rare. Here, we report an experiment in which third-grade children initially read a brief encyclopaedic text twice and then either performed a cued recall test on selected target contents of the text or reread the same text twice. A final four-choice memory test about the text's contents was administered to all children after seven days as well as tests measuring their IQ and reading skills. In the final four-choice memory test, children who took the initial cued recall test showed significantly greater performance in comparison with children who restudied the text twice (their responses were 87 and 53% correct, respectively). The results suggest that cued recall tests can elicit very robust testing effects in young children, even when complex, educationally relevant materials are used as stimuli.

Keywords: testing effect; test-enhanced learning; memory; learning; education

Introduction

An extensive body of research has shown that encoding processes are critical for the successful storage of long-term memories (e.g. Craik & Lockhart, 1972). However, recent research has highlighted the importance of retrieval processes in the successful retention of such memories (Butler, 2010; Butler & Roediger, 2008; Carpenter, Pashler, & Cepeda, 2009; Halamish & Bjork, 2011; Karpicke & Roediger, 2007; Roediger & Karpicke, 2006). Studies demonstrating this fact typically involve experiments in which the impact of retrieval is directly compared with the impact of restudy on the retention of memories. In such studies, a series of items are studied in an initial stage, after which the participants either study these items again or retrieve them from memory. The key finding provided by such experiments is that although restudied items may be more successfully remembered than retrieved items after brief intervals, retrieved items are more likely to be successfully remembered than restudied items at later stages. That is, retrieval enhances the retention of long-term memories in comparison with restudy, which is a phenomenon frequently termed *testing effect* or *test-enhanced learning* (Karpicke, 2012; Roediger & Karpicke, 2006).

Several recent studies have replicated this effect (for a review, see Roediger & Butler, 2011). These studies have shown that it is exhibited by individuals in different

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age ranges (Tse, Balota, & Roediger, 2010), after various study-test intervals (Roediger & Butler, 2011), and when words (Zaromb & Roediger, 2010), texts (Roediger & Karpicke, 2006), and spatial information (Carpenter & Kelly, 2012) are used as stimuli. Although several studies have investigated the testing effect in college students, fewer have investigated it in children within laboratory settings (Bouwmeester & Verkoeijen, 2011; Goossens, Camp, Verkoeijen, & Tabbers, 2014; Roher, Taylor, & Sholar, 2010), or using educational materials within educational settings (Carpenter et al., 2009; McDaniel, Agarwal, Huelser, McDermott, & Roediger, 2011; Roediger, Agarwal, McDaniel, & McDermott, 2011).

In laboratory settings, Roher et al. (2010) have demonstrated that fourth and fifth graders exhibit testing effects in map-learning tasks that require the transfer of knowledge on a final test, while Bouwmeester and Verkoeijen (2011) used a series of Deese–Roediger–McDermott lists (Deese, 1959; Roediger & McDermott, 1995) as stimuli and showed that even seven-year-old children can exhibit testing effects. Similarly, Goossens et al. (2014) reported an experiment in which children either restudied new vocabulary several times or practised the retrieval of such vocabulary while also restudying it several times. This manipulation showed that retrieval practice associated with restudy was more effective than restudy alone in enhancing third-grade children's cued recall of new vocabulary in a later test.

Even though the studies above showed that the testing effect can occur in young children, the laboratory tasks they employed do not resemble the typical tasks encountered in educational settings. In the experiments reported by Roher et al. (2010), for instance, the task of learning the name of locations on a map was conducted at a very fast pace (about 6 s per trial), which contrasts with typical (usually much slower) educational practices. In the studies of Bouwmeester and Verkoeijen (2011) and Goossens et al. (2014), alternatively, children learned or were tested with lists of individual or paired words, tasks that are also not typically employed in the classroom.

In an attempt to verify whether the testing effect exhibited in laboratory settings can be replicated in educational environments with educational materials, Carpenter et al. (2009) demonstrated that eighth-grade students showed greater retention of materials taught during their regular courses when these materials were restudied through questions followed by feedback, in contrast to when these materials were only reread. Similarly, Lipko-Speed, Dunlosky, and Rawson (2014) showed that retrieval practice followed by feedback enhanced the learning of science concepts by fifth graders, whereas Roediger et al. (2011), McDaniel, Thomas, Agarwal, McDermott, and Roediger (2013), McDermott, Agarwal, D'Antonio, Roediger, and McDaniel (2014) and McDaniel et al. (2011) showed that sixth-, seventh- and eighth-grade children significantly improved their long-term retention of class materials when quizzes consisting of multiple-choice questions were periodically administered in the classroom.

Although these studies were initial attempts to implement and examine the testing effect in educational environments, they provide strong evidence that this can be an effective approach to improve learning in such environments. However, further studies are still needed to examine whether this effect is elicited by younger children when educational materials are used. While experimental data show that children as young as 7 years of age can exhibit testing effects (Bouwmeester & Verkoeijen, 2011), we are not aware of studies demonstrating testing effects using materials typically employed in educational settings for children at such early ages.

Another issue that remains in need of further investigation is the potential association between individual differences and memory performance after retrieval practice. As shown by prior studies, there is a clear association between IQ and performance on several memory tasks (McKoon & Ratcliff, 2012; Ratcliff, Thapar, & McKoon, 2010). However, the potential role of IQ in the test-enhanced learning effect is yet to be investigated. Similarly, even though most experiments investigating testing effects employed words or texts as stimuli (e.g. Roediger & Karpicke, 2006), we are not aware of experiments examining potential associations between reading skills and testing effect. This issue is especially important in studies with young children, since they are still developing literacy (Chall, 1983) and therefore may present substantial variability in the processing of verbal stimuli (Spira, Bracken, & Fischel, 2005).

In sum, to investigate the testing effect experimentally in children, using educationally relevant materials, third-grade children initially read a brief text and subsequently performed a cued recall test on target facts from the text or simply restudied the text twice. Their long-term retention of the content of the text was tested one week later in a four-choice recognition task. Even though we used materials that resembled those typically used at class, the present design provided more rigorous control of the information encoded by the participants than prior studies employing such materials with children (e.g. Roediger et al., 2011). Furthermore, we also assessed IQ and reading skills to investigate potential associations between these scores and the effects of testing.

Based on prior findings, we expected that the group who performed the cued recall test instead of restudying the text would perform better in the later memory test. We also expected that individuals with higher IQ and higher reading skills would show higher performance in the memory tests. Nonetheless, we refrain ourselves from advancing predictions regarding potential associations between the measures of IQ and reading skills with the testing effect, since we are not aware of prior empirical data showing such associations.

Methods

Participants

The participants were 69 children with ages ranging from 8 years and 10 months to 10 years and 9 months (mean age = 10 years and 1 month). All children were enrolled in the third grade of four Brazilian elementary schools located in southern Brazil and were from low-income and medium-income families. Forty children (mean age = 10 years and 1 month, SD = 5.30 months, 18 females) were assigned to the test group, and 29 (mean age = 10 years and 2 months, SD = 4.81 months, 22 females) were assigned to the restudy group. Children in the same classroom were assigned to the same experimental condition. Twenty-one additional children who were initially enrolled in the study were excluded due to not showing up for the second session of the experiment. Informed consent was obtained in accordance with the institutional review board of the Pontifical Catholic University of Rio Grande do Sul, and both the children and their parents filled in signed consent forms.

Materials and stimuli

The study material was a text containing 321 words about the sun, which was adapted from a Wikipedia website in Brazilian Portuguese (http://pt.wikipedia.org/wiki/Sol). To assess whether the text was appropriate for third-grade students, three elementary school teachers and two biology middle-school teachers, as well as a linguist formally trained in applied linguistics assessed the appropriateness of the language, vocabulary, formal structure, and length of the text for third-grade students. The assessment of each of these topics was performed using a four-point scale (from 'inappropriate' to 'appropriate'). The participants' teachers also reported on whether the topic covered in the text (the sun) was ever studied in class using a four-point scale ('never studied' to 'intensely studied'). The text was considered 'appropriate' in terms of language, vocabulary, formal structure, and length by all independent judges. The participants' teachers classified the topic covered by the text as having 'never [been] studied' in class.

Twenty 'target facts' were selected from the text and used as probes for the memory tests. Each target fact consisted of a word within a sentence of the text, while each sentence was used as a cue in the cued recall and recognition tests. Examples of target facts and sentences used as cues are 'The word sun derives from the Latin word *solis*', 'The sun is the central star of our *solar* system' and 'In our solar system, there are also planets, satellites, and other small celestial bodies, such as comets and *meteors*'. In these examples, the words in italic font were used as target facts, whereas the sentences (without the target facts) were used as cues.

The buffer task, which was administered just after all of the participants read the text twice, consisted of performing calculations using the four basic mathematical operations (addition, subtraction, multiplication and division). The grade appropriateness of the particular operations was also assessed by independent judges (three elementary school teachers), who unanimously considered the task to be 'appropriate'. The cued recall task involving the 20 selected target facts (e.g. 'the superficial layer of the sun is called _____') was administered to the participants of the test group, whereas the participants of the restudy group just read the text twice again. No feedback about the correctness of the children's responses was provided during the cued recall task. After seven days, all participants performed a four-choice recognition task with the same target facts (e.g. 'The superficial layer of the sun is called _____ (a) gas, (b) energy, (c) glucose, (d) photosphere').

In order to assess the children's reading skills, the reading comprehension test *Avaliação da Compreensão de Leitura Textual* (Salles & Parente, 2004) was administered to all participants. It is validated for Brazilian Portuguese speaking children and involves responding to 10 multiple-choice questions that assess comprehension and interpretation of a narrative text. Intelligence was measured by a version of the Raven Coloured Progressive Matrices (RCPM) that was translated and validated for Brazilian samples (Angelini, Alves, Custódio, Duarte, & Duarte, 1999). It classifies children's intelligence within five levels (I–V) based on percentile measures. Thus, level I denotes the 95th percentile or above, suggesting superior intelligence, and V denotes the fifth percentile or below, suggesting mental deficiency.

Procedures

The memory task involved two sessions separated by an interval of seven days. All stages of the memory task were performed individually by the participants in their own classroom during school hours and in the presence of their teachers and classmates. Except for the RCPM, all of the tasks were performed and registered in individual booklets distributed to the students at the beginning of each session. During the first session, all of the participants read the text 'The Sun' twice as soon as the experiment started. Following this, the participants took the mathematical operations buffer task for three minutes. Next, the participants of the test group performed a cued recall task, whereas the participants of the restudy group read the text twice again. Seven days later, all participants performed a four-choice recognition task with the target facts. The reading skills test *Avaliação da Compreensão de Leitura Textual* and the RCPM were administered by the experimenter in the days following the final session of the memory task.

Results

Both the test and restudy group showed equivalent performance in reading comprehension, as assessed by the *Avaliação da Compreensão de Leitura Textual* (Salles & Parente, 2004). The mean proportion of correct responses was .91 (SD = .1) for the test group and .92 (SD = .09) for the restudy group, $t < 1$. Intelligence was assessed by the raw scores of the RCPM (maximum score = 36). Even though both groups showed above average intelligence, with both groups showing mean intelligence scores within level II, the test group showed a poorer performance than the restudy group in this measure ($M = 27.92$, SD = 5.73, and $M = 31.31$, SD = 3.0 respectively), yielding a statistically significant difference between the groups' mean scores, $t(67) = 2.93$; $p < .005$, $d = .72$. Potential associations between the individual scores on these tests and memory performance will be explored below.

Memory performance

For the analysis of memory performance, each correctly retrieved target fact was tabulated as 1 (maximum performance = 20). In the initial test (cued recall), which was administered within the test group only, children recalled an average of 13.1 (SD = 3.44) target facts, corresponding to 65.5% of all selected target facts. In the final test (the four-choice recognition test), which was conducted seven days after the first session and administered to participants from both the test and the restudy group, children from the test group recognised an average of 17.4 (SD = 1.81) target facts (87%), whereas children from the restudy group recognised an average of 10.6 (SD = 2.98) target facts (53%), yielding a robust between-group statistical difference, $t(67) = 11.73$; $p < .001$, $d = 2.87$. This difference is strong evidence that the testing effect can be exhibited by third-grade children in educational settings when educational materials are used.

To investigate the potential associations between intelligence and memory performance, we conducted a correlation analysis between the participants' IQ scores, as measured by the RCPM (Angelini et al., 1999), and performance in both the cued recall (test-group only) and the multiple-choice recognition tests. Performance in the cued recall test was positively correlated with IQ, $r = .46$, $p < .05$, whereas significant correlations between IQ and recognition were shown in neither the restudy

group nor the test group. Thus, even though no association is apparent between later recognition performance and intelligence, higher intelligence children showed greater performance on cued recall relative to lower intelligence children.

In order to evaluate memory performance for children with different reading skills, we conducted correlation analyses between their scores in the *Avaliação da Compreensão de Leitura Textual* test (Salles & Parente, 2004) and their performance on the cued recall and multiple choice recognition tests. These analyses showed no correlation between reading skills and memory performance in the cued recall test. However, an interesting pattern emerged from the analysis of recognition performance. Namely, while reading skills were positively correlated with recognition performance for the restudy group, $r = .37$, $p < .05$, there were no significant correlations between these variables for the test group ($r = -.06$). A tentative interpretation for this finding will be presented in the Discussion section.

Discussion

The experiment reported here investigated the testing effect in third-grade children using material that is typically encountered in educational settings. The material consisted of a brief text, which was read by all of the participants, who then either performed a cued recall test on selected target facts of the text or restudied the text twice. A final multiple-choice recognition test administered seven days later showed that children who performed the initial cued recall test exhibited greater retention of the target facts than children who restudied the text, even though the latter group restudied the text twice.

In line with prior studies (e.g. Butler, 2010; Butler & Roediger, 2008; Carpenter et al., 2009; Halamish & Bjork, 2011; Karpicke & Roediger, 2007; Roediger & Karpicke, 2006), and especially with studies recruiting children as participants (Carpenter et al., 2009; McDaniel et al., 2011; Roediger et al., 2011), we provide strong evidence that third-grade children can exhibit the testing effect in educational settings when educationally relevant materials are used. As in studies employing experimental designs (Bouwmeester & Verkoeijen, 2011; Goossens et al., 2014; Roher et al., 2010), our manipulation granted us substantial control over the encoded material, since here all participants performed the same encoding task.

All participants in the current study were enrolled in the third grade and had an overall mean age of 10 years and 1 month. Although Lipko-Speed et al. (2014) and Roediger et al. (2011) have not reported their participants' mean ages, their participants were enrolled in the fifth and sixth grades. Thus, despite potential differences in the school calendar (e.g. we recruited our participants towards the end of the school year) and international differences in the typical age range assigned to each grade, our sample is considerably younger than the samples recruited in those studies. Thus, the present study represents an important step in broadening our knowledge of the age range in which the testing effect is exhibited when complex, educational-like materials are used. Further studies would be useful to expand this knowledge by determining the minimum age in which testing effects are exhibited. Such a finding could provide valuable data for future applications of the testing effect in actual educational settings.

A significant positive correlation was found between reading skills and memory performance in the later recognition test for the restudy group, an effect that was clearly absent for the test group. One possible account for this finding is that the

restudy group had read the text four times instead of twice, as the test group did. Consequently, children with higher reading skills would naturally benefit more from reading the same text over and over again, in comparison to children with lower reading skills. Future translational research should take notice of this issue and perhaps investigate how this can best be applied to improve the learning of children with diverse reading skills.

A positive correlation was also found between IQ and performance in the cued recall test, although no correlation was found between IQ and performance in the later recognition test. This pattern is likely due to differences in the cognitive demands involved in each test. Whereas recognition tests can be performed based mainly on familiarity processes, cued recall is frequently associated with recollective processes (Yonelinas, 2002). Future research could use cued recall tests to verify whether such correlations remain significant after long intervals (e.g. seven days) and, more importantly, whether such correlations vary according to whether the participants previously performed a test or restudied the stimulus materials.

In contrast to prior studies investigating the testing effect in children with educationally relevant materials (Carpenter et al., 2009; McDaniel et al., 2011; Roediger et al., 2011), we did not provide feedback regarding the correctness of the participants' responses in the current experiment. Prior findings have demonstrated that providing feedback when participants are retrieving studied information increases the testing effect relative to when participants do not receive feedback (McDaniel, Roediger, & McDermott, 2007; Roediger & Karpicke, 2006). On the other hand, the robust effects encountered here suggest that feedback is not crucial for eliciting strong testing effects, and tests without feedback (as the paper-and-pencil cued recall test administered here) might be powerful tools to improve learning in educational settings.

The strong testing effect reported here is in line with the idea that effort during successful retrieval can be critical for increasing later retention of the retrieved material. According to this hypothesis, difficult retrievals are better for memory retention than easy retrievals (Pyc & Rawson, 2009). Thus, since we used a cued recall test here – a task that is usually more effortful than multiple choice recognition tasks (McDaniel et al., 2007) – the high effort exerted when the participants were retrieving the selected target facts might have been a critical catalyst of the current effect. Future research envisioning applications of the testing effect in educational settings could profit by using protocols in which effortful retrieval is manipulated. Furthermore, further research would be useful to determine whether, in educational settings, cued recall tests are more effective in increasing retention than multiple-choice questions followed by feedback (e.g. Roediger et al., 2011).

In sum, we showed that cued retrieval is highly beneficial for enhancing the memory retention of educationally relevant materials in children. In addition to findings from prior studies investigating this effect in children (Carpenter et al., 2009; McDaniel et al., 2011; Roediger et al., 2011), the present study reveals the robustness of this effect for children who are still enrolled in the third grade.

Conflict of interest

We certify that there is no conflict of interest with any financial organisation regarding the material discussed in this manuscript.

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